

Electrical Machine- II (BT EE403A)



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Unit-IV

Synchronous Machine

- ❖ Reasons of Parallel Operation
- ❖ Necessary Conditions for Parallel Operation of the Alternator
- ❖ Parallel Operation of two Generators

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Reasons of Parallel Operation

Alternators are operated in parallel for the following reasons:

- Several alternators can supply a bigger load than a single alternator.
- One or more alternators may shut down during the period of light loads. Thus, the remaining alternator operates at near or full load with greater efficiency.
- When one machine is taken out of service for its scheduled maintenance and inspection, the remaining machines maintain the continuity of the supply.
- If there is a breakdown of the generator, there is no interruption of the power supply.
- Number of machines can be added without disturbing the initial installation according to the requirement to fulfill the increasing future demand of the load.
- Parallel operation of the alternator, reduces the operating cost and the cost of energy generation.
- It ensures the greater security of supply and enables overall economic generation.

Necessary Conditions for Parallel Operation of the Alternator

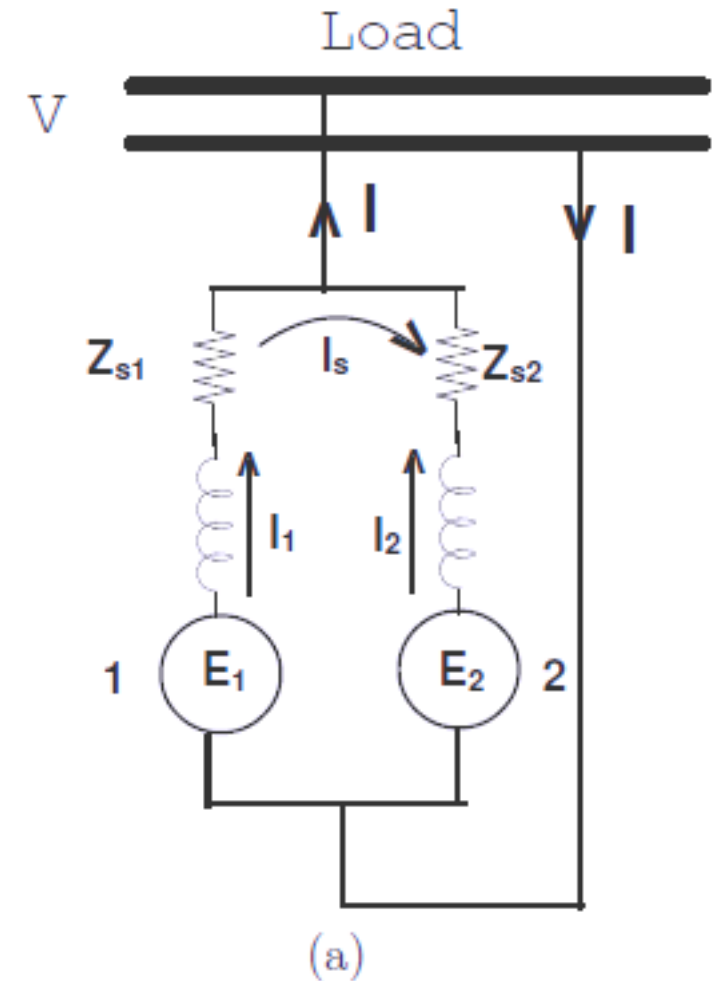
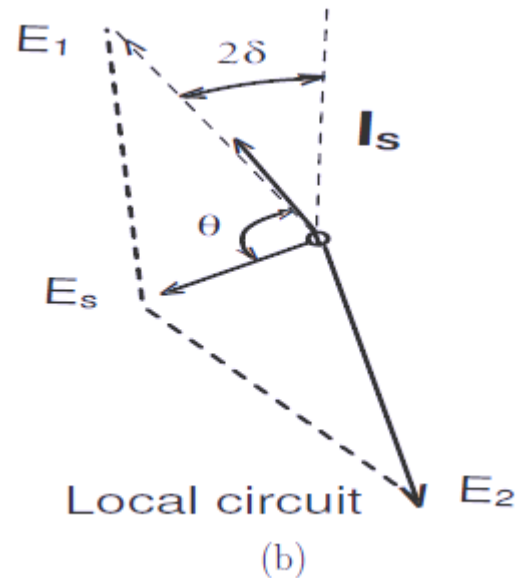
Most synchronous machines will operate in parallel with other synchronous machines. The process of connecting one machine in parallel with another machine or with an Infinite Bus bar system is known as Synchronizing. The machine carrying load is known as Running Machines while the alternator which is to be connected in parallel with the system is known as the Incoming machine.

The following condition should be satisfied for parallel operation are as follows:-

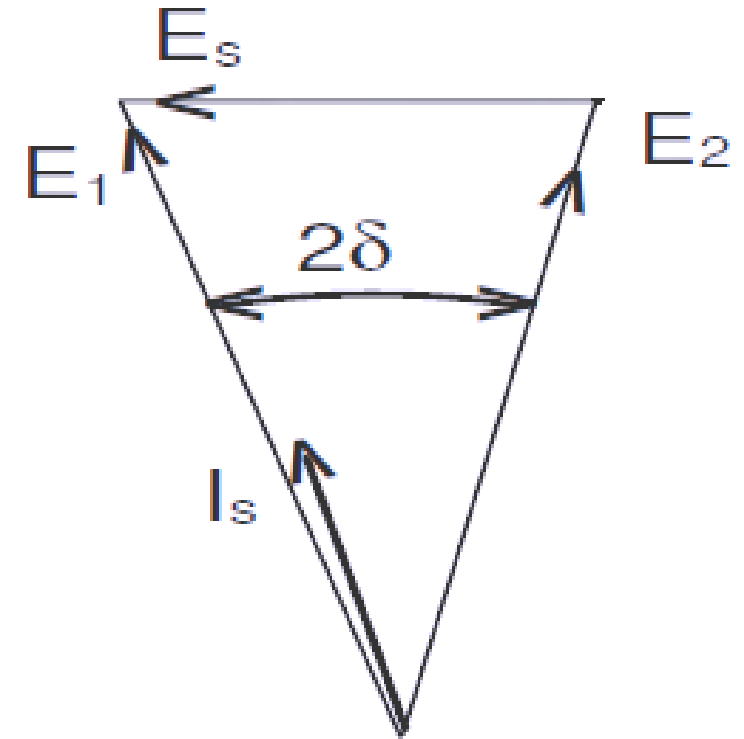
- The phase sequence of the Bus bar voltages and the incoming machine voltage must be the same.
- The Bus bar voltages and the incoming machine terminal voltage must be in phase.
- The terminal voltage of the incoming machine and the alternator which is to be connected in parallel or with the Bus bar voltage should be equal.
- The frequency of the generated voltage of the incoming machine and the frequency of the voltage of the Bus bar should be equal.

Parallel Operation of two Generators

When two synchronous generators are connected in parallel, they have an inherent tendency to remain in step, on account of the changes produced in their armature currents by a divergence of phase. Consider identical machines 1 and 2, Fig (a) in parallel and working on to the same load. With respect to the load, their e.m.f.s are normally in phase: with respect to the local circuit formed by the two armature windings, however, their e.m.f.s are in phase-opposition, Fig (b).



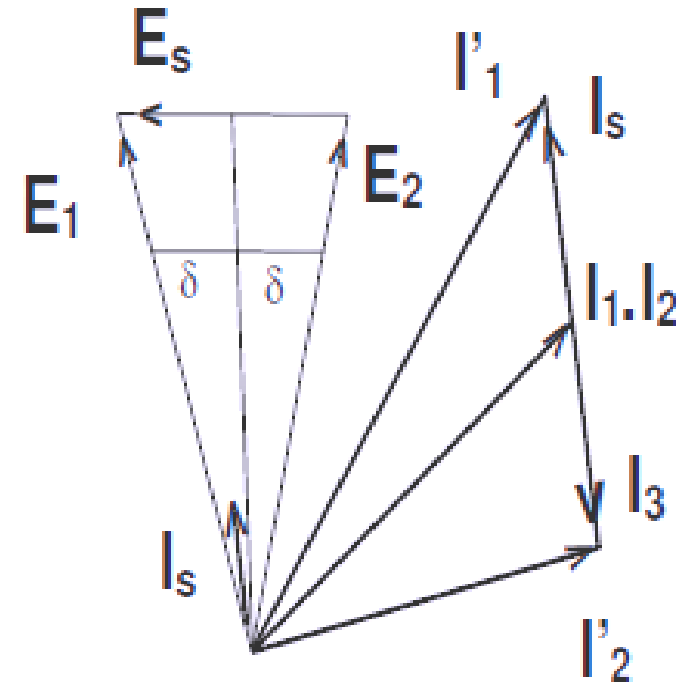
Suppose there to be no external load. If machine 1 for some reason accelerates, its e.m.f. will draw ahead of that of machine 2. The resulting phase difference 2δ causes e.m.fs to lose phase-opposition in the local circuit so that there is in effect a local e.m.f E_s which will circulate a current I_s in the local circuit of the two armatures. The current I_s flows in the synchronous impedance of the two machines together, so that it lags $\theta = \arctan(x_s/r) \approx 90^\circ$ on E_s on account of the preponderance of reactance in Z_s ; I_s therefore flows out of machine 1 nearly in phase with the e.m.f., and enters 2 in opposition to the e.m.f. Consequently machine 1 produces a power $P_s \sim E_t \cdot I_s$ as a generator, and supplies it (I^2R losses excepted) to 2 as a synchronous motor.



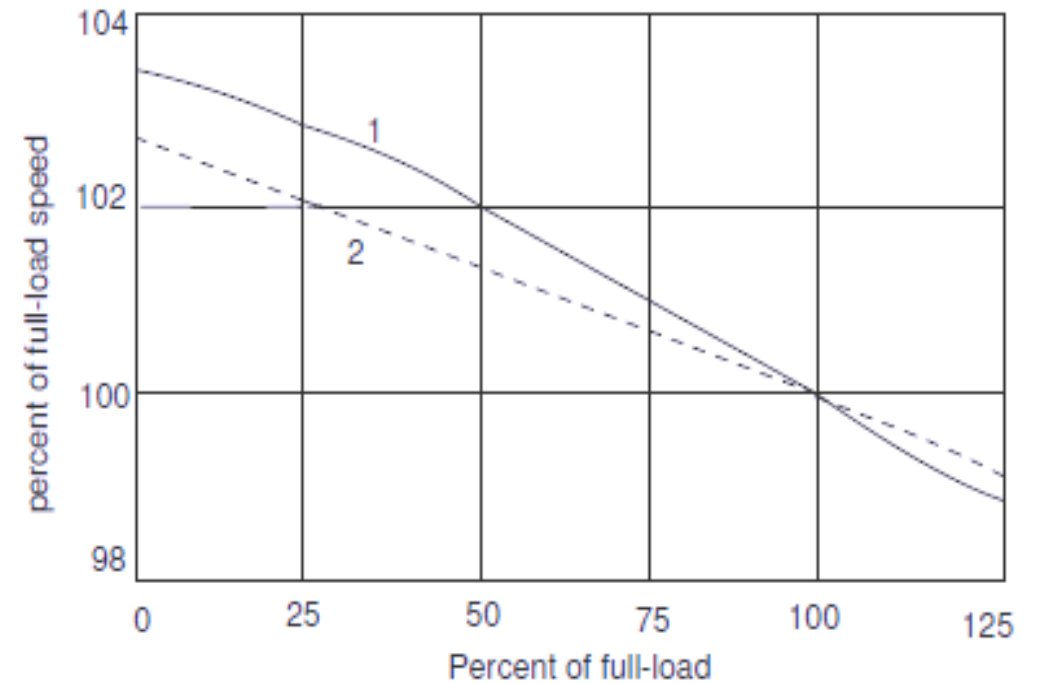
➤The synchronizing power P_s tends to retard the faster machine 1 and accelerate the slower, 2, pulling the two back into step. Within the limits of maximum power, therefore, it is not possible to destroy the synchronous running of two synchronous generators in parallel, for a divergence of their angular positions results in the production of synchronizing power, which loads the forward machine and accelerates the backward machine to return the two to synchronous running.

➤The development of synchronizing power depends on the fact that the armature impedance is preponderating reactive. If it were not, the machines could not operate stably in parallel: for the circulating current I_s would be almost in phase- quadrature with the generated e.m.f.'s, and would not contribute any power to slow the faster or speed up the slower machine.

When both machines are equally loaded on to an external circuit, the synchronizing power is developed in the same way as on no load, the effect being to reduce the load of the slower machine at the same time as that of the faster machine is increased.



The governors of prime movers are usually arranged so that a reduction of the speed of the prime mover is necessary for the increase of the power developed. Unless the governor speed/load characteristics are identical the machines can never share the total load in accordance with their ratings. The governor characteristics take the form shown in fig. If the two are not the same, the load will be shared in accordance with the relative load values at the running speed, for synchronous machines must necessarily run at identical speeds.



Governor Characteristic